

**M2 Conditional probability
A level only**

Name: _____

Class: _____

Date: _____

Time: **401 minutes**

Marks: **324 marks**

Comments:

EXAM PAPERS PRACTICE

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Q1.

Of the patients in an accident unit, 30% have a sports injury.

Past records for the accident unit suggest that:

a patient with a sports injury has a probability of 0.2 of being admitted to hospital;

a patient who does **not** have a sports injury has a probability of 0.4 of being admitted to hospital.

A patient is chosen at random from those in the accident unit.

- (a) (i) Write down the probability that the chosen patient has a sports injury. (1)
- (ii) Find the probability that the chosen patient has a sports injury and is admitted to hospital. (1)
- (iii) Show that the probability that the chosen patient is admitted to hospital is 0.34. (2)
- (b) Given that the chosen patient is admitted to hospital, find the conditional probability that the patient has a sports injury. (3)
- (Total 7 marks)**

Q2.

A customer goes into a store to buy a refrigerator and a microwave. From past experience it is known that 10% of the refrigerators and 5% of the microwaves will be found to be defective when tested. The customer chooses one refrigerator and one microwave at random and the items are tested.

- (a) Find the probability that:
- (i) both items are found to be defective; (2)
- (ii) neither item is found to be defective; (2)
- (iii) exactly one of the items is found to be defective. (2)
- (b) Given that exactly one of the items is found to be defective, find the probability that it is the refrigerator. (3)
- (Total 9 marks)**

Q3.

A football club employs 30 players, of whom 13 have British nationality. The following table shows how many of each type of player are employed by the club.

	Goalkeepers	Defenders	Midfielders	Attackers
British	2	4	5	2
Other	2	5	7	3

One player is chosen at random from the 30 players.

(a) Find the probability that the chosen player is:

(i) British;

(1)

(ii) a goalkeeper;

(1)

(iii) British and a goalkeeper;

(1)

(iv) a defender or a midfielder;

(1)

(v) British or an attacker but **not** both.

(2)

(b) (i) Given that the chosen player is British, find the probability that he is a goalkeeper.

(2)

(ii) State, giving a reason, whether the events
 'the chosen player is British' and
 'the chosen player is a goalkeeper'
 are independent.

(2)

(Total 10 marks)

Q4.

Jane is an athlete in training for a major event. Every evening she does either a 5-mile run or a 10-mile run. On a given day Jane does a full day's work with probability $\frac{2}{3}$.

When she has done a full day's work, the probability that she does a 10-mile run is $\frac{2}{5}$.

When she has done a **not** full day's work, the probability that she does a 10-mile run is $\frac{4}{5}$.

- (a) For a random day, calculate the probability that:
- (i) she does a full day's work and a 10-mile run; (2)
 - (ii) she does **not** do a full day's work but she does a 10-mile run; (2)
 - (iii) she does a 10-mile run. (1)
- (b) Given that she does a 10-mile run, find the conditional probability that she has done a full day's work. (2)
- (Total 7 marks)

Q5.

A boy receives a school report card each week.

He is given a special treat whenever his report indicates "Good behaviour" **and** "Excellent homework".

His behaviour is good with probability 0.6.

When his behaviour is good, he has a probability of 0.8 of doing excellent homework.

When his behaviour is **not** good, his probability of doing excellent homework is only 0.5.

- (a) For a random week, calculate the probability that:
- (i) he will be given a special treat; (2)
 - (ii) his homework will be excellent but he will **not** be given a special treat; (2)
 - (iii) his homework will be excellent. (1)
- (b) Given that his homework is excellent, calculate the conditional probability that he is **not** given a special treat. (3)
- (Total 8 marks)

Q6.

Two letters are posted on the same day.

Letter A is sent by first-class post and has a probability of 0.9 of being delivered the next day.

Letter B is sent by second-class post and has a probability of only 0.3 of being delivered the next day.

(a) Find the probability that:

(i) **both** letters are delivered the next day; (2)

(ii) **neither** letter is delivered the next day; (2)

(iii) **at least one** of the letters is delivered the next day. (2)

(b) Given that at least one of the letters is delivered the next day, find the probability that letter A is delivered the next day.

(3)

(Total 9 marks)

Q7.

It is known that 80% of the population are inoculated against a particular disease. Of those who are inoculated, 90% are then immune to the disease, whereas only 5% of those not inoculated are immune.

(a) Use a tree diagram, or otherwise, to calculate the probability that a randomly selected member of the population is immune. (5)

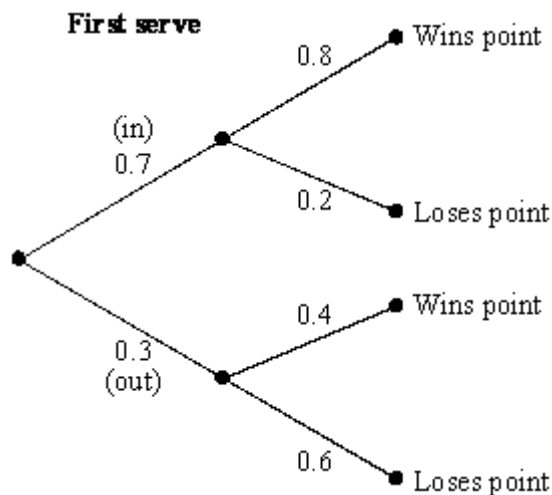
(b) Given that a member of the population is immune, find the conditional probability that he or she has been inoculated. (3)

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(Total 8 marks)

Q8.

A tennis player is allowed two serves. The probability that a particular player's first serve will be in is 0.7. She has probability 0.8 of winning the point when her first serve is in. Otherwise, her probability of winning the point is only 0.4. This information is shown in the tree diagram.



- (a) Find the probability that she wins a point when she is serving.

(3)

- (b) Given that she does win this point, find the probability that her first serve was in.

(3)

(Total 6 marks)

Q9.

In a group of students 60% are female and 40% are male. A third of the female students study Spanish but only a quarter of the male students study Spanish.

A student is chosen at random from the group.

- (a) (i) Show that the probability that the chosen student is female and studies Spanish is 0.2.

(2)

- (ii) Calculate the probability that the chosen student studies Spanish.

(2)

- (b) Given that the chosen student **does** study Spanish, calculate the conditional probability that the chosen student is female.

(3)

(Total 7 marks)

Q10.

A school employs 75 teachers. The following table summarises their length of service at the school, classified by gender.

	Less than 3 years	3 years to 8 years	More than 8 years
Female	12	20	13
Male	8	15	7

- (a) Find the probability that a randomly selected teacher:
- (i) is female;
 - (ii) is female, given that the teacher has more than 8 years service;
 - (iii) is female, given that the teacher has less than 3 years service.
- (4)**
- (b) State, giving a reason, whether or not the event of selecting a female teacher is independent of the event of selecting a teacher with less than 3 years service.
- (2)**
- (c) Define an event which is mutually exclusive to the event of selecting a female teacher.
- (2)**
- (d) Three teachers are selected at random without replacement. Find the probability that all three are:
- (i) females with less than 3 years service;
 - (ii) of the same gender.
- (5)**
- (Total 13 marks)**

Q11.

The probability that a climber chooses to attempt to ascend a mountain by the North Ridge is 0.5, by the West Face is 0.3, and by the East Face is 0.2.

The probability that a climber has a fall in an attempted ascent on the North Ridge is 0.2, on the West Face is 0.4, and on the East Face is 0.1.

- (a) Find the probability that a climber chooses to ascend the mountain by the North Ridge and has a fall.
- (2)**
- (b) Find the probability that a climber has a fall in a single attempted ascent of the mountain.
- (3)**
- (c) Given that a climber is known to have had a fall, find the probability that the climber made the attempted ascent by the North Ridge.
- (3)**
- (d) It may be assumed that a climber who does not have a fall reaches the summit.
- Given that a climber reaches the summit, find the probability that the climber chose to ascend the mountain by the North Ridge.
- (3)**
- (Total 11 marks)**

Q12.

The population of a country is suffering from an epidemic of a serious disease.

- (a) A hospital doctor is consulted by 250 people suffering from symptoms of this disease. Of the 250 people, 105 are male, of whom 55 have the disease. Of the females, 65 have the disease.

A person is selected at random from the 250 people who consulted the hospital doctor.

The following events are defined:

F is the event "the person is female";
 S is the event "the person has the disease";
 S^c is the event "not S ".

Find:

- (i) $P(F)$;
 (ii) $P(F \cap S^c)$;
 (iii) $P(S)$;
 (iv) $P(F | S)$.

(5)

- (b) The probability that a person selected at random from a large population has the disease is 0.2.

- (i) Four people are selected at random from a large population. Find the probability that at least one of these four people has the disease.

(2)

- (ii) Of the people suffering from this disease, 90% give a positive reaction to the test designed to detect the disease, whereas 15% of the people who are not suffering from the disease also give a positive reaction to the test.

Calculate the probability that a person who gives a positive reaction to the test does not have the disease.

(4)

(Total 11 marks)

Q13.

A co-operative owns and runs a shop specialising in organic vegetables. The number of members of the co-operative, classified by age and sex, is given in the table below.

	Age (years)		
	Under 30	30-49	50 and over
Female	3	6	4

Male	2	8	5
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- (a) A member is selected at random to represent the co-operative at a meeting. Find the probability that the member selected is:
- (i) female and aged under 30; (1)
 - (ii) aged under 30; (1)
 - (iii) aged under 30, given that the member is female. (2)
- (b) Two members are selected at random to represent the co-operative at another meeting. Find the probability that:
- (i) both are females aged 30–49; (2)
 - (ii) one is male and one is female. (2)
- (c) R denotes the event that the member selected in part (a) is female.
 S denotes the event that the member selected in part (a) is aged under 30.
 T denotes the event that the member selected in part (a) is aged 30–49.
- (i) Write down two of the events R, S and T which are mutually exclusive. (1)
 - (ii) State whether or not events R and S are independent. Justify your answer. (2)
- (Total 11 marks)**

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Q14.

In a large apartment building, a random sample of residents was asked their opinions regarding the proposed maintenance charges for the following year. Each resident occupies either a one-bedroomed or a two-bedroomed apartment.

They were each asked to indicate whether they felt the level of the charges was about right, slightly too high or excessive.

The following table summarises their replies.

Reply Apartment	About right	Slightly too high	Excessive
One-bedroomed	18	46	23
Two-bedroomed	35	39	9

A resident is selected at random.

B is the event 'resident occupies a one-bedroomed apartment'.

R is the event 'resident replied "about right"'.

H is the event 'resident replied "excessive"'.

B' is the event 'not B '.

(a) Find:

(i) $P(B)$; (1)

(ii) $P(B \cap R)$; (1)

(iii) $P(H \cup B)$; (2)

(iv) $P(H | B')$. (3)

(b) Describe in words, as simply as possible, the event $(H \cap B')$. (2)

(Total 9 marks)

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Q15.

A group of 50 female and 30 male students is questioned about their eating habits.

(a) Of the females, 40 per cent do not eat breakfast and, of the males, 10 per cent do not eat breakfast.

Find the probability that a student, selected at random from the group:

(i) does not eat breakfast; (3)

(ii) is female, given that the student does not eat breakfast. (2)

(b) Of all the students in the group, 10 per cent eat a hot breakfast, 20 per cent eat a hot lunch and 25 per cent either eat a hot breakfast or eat a hot lunch or eat both.

Find the probability that a student, selected at random from the group:

- (i) eats both a hot breakfast and a hot lunch; (2)
 - (ii) eats a hot breakfast, given that the student eats a hot lunch. (2)
- (Total 9 marks)

Q16.

A newspaper published an article concerning a proposed European law which would force motorists to pay compensation and damages in all accidents with cyclists, regardless of who is to blame. The newspaper received 150 letters in response to the article. The following table shows the number of letters received, classified by the writers' usual forms of transport and their attitudes to the proposal.

		Attitude to the proposal		
		For	Against	Neutral
Usual form of transport	Car	3	67	9
	Bicycle	27	5	10
	Other	8	12	9

- (a) The editor selects a letter at random to consider for publication. Find the probability that the letter selected is:
 - (i) for the proposal; (3)
 - (ii) for the proposal, given that the writer usually travels by car. (3)
- (b) The editor selects three letters at random, without replacement. Find the probability that one letter is for the proposal, one against and one neutral. (4)
- (c) The proportions of letters suitable for publication were:
 - 0.50 of those which were for the proposal;
 - 0.25 of those which were against the proposal;
 - 0.75 of those which were neutral.
 - (i) Find the probability that a randomly selected letter is for the proposal and suitable for publication. (2)
 - (ii) Find the probability that a randomly selected letter is suitable for publication. (3)
 - (iii) Find the probability that at least three letters will have to be selected (at

random, without replacement) before a letter is found which is suitable for publication from a writer who is for the proposal.

(3)

(Total 15 marks)

Q17.

On a visit to a theme park, the probabilities that a visitor will go on the Monorail, the Big Mountain ride and the Fearsome ride are 0.65, 0.52 and 0.38 respectively.

The probability that a visitor will go on the Big Mountain ride and the Fearsome ride is 0.25.

The probability that a visitor will go on the Monorail and the Big Mountain ride is 0.30. The probability that a visitor will go on the Fearsome ride, given that they go on the Monorail is 0.46.

Find the probability that a randomly selected visitor to this theme park will go on:

(a) the Monorail and the Fearsome ride;

(2)

(b) the Monorail or the Big Mountain ride or both;

(2)

(c) neither the Big Mountain ride nor the Fearsome ride;

(3)

(d) the Monorail, given that they go on the Fearsome ride.

(3)

(Total 10 marks)

Q18.

The age and the blood pressure status for each adult male in a randomly selected sample are summarised in the following table.

Blood pressure status \ Age Range	25 – 34	35 – 54	55 – 74
Normal (untreated)	52	47	28
High (untreated)	4	7	18
High (treated)	1	4	13

One of the adult males is selected at random.

H is the event 'the male selected has high blood pressure'.

R is the event 'the male selected is aged 55–74'.

S is the event 'the male selected is aged 25–34'.

T is the event 'the male selected is being treated'.

R' is the event 'not R '.
 H' is the event 'not H '.

- (a) Find:
- (i) $P(S)$; (1)
 - (ii) $P(T)$; (1)
 - (iii) $P(R \cup T)$; (2)
 - (iv) $P(H \cap R')$; (2)
 - (v) $P(S|T)$; (3)
 - (vi) $P(R|H')$. (3)
- (b) Find the probability of the event $(R \cap H \cap T)$.
 Define this event in context as simply as possible. (3)

(Total 15 marks)

Q19.

When Gordon and Louise play a board game, the probability that Gordon wins is 0.6 and the probability that Louise wins is 0.4. They agree to play a series of games. The winner of the series will be the first player to win three games. The result of each game is independent of the result of any other game.

Find the probability that:

- (a) Gordon wins the series by three games to nil; (2)
- (b) more than three games are necessary to decide who wins the series; (3)
- (c) Louise wins the series by three games to one, given that she wins the first game; (3)
- (d) Louise wins the series, given that the final score is three games to two; (2)
- (e) Gordon wins the series, given that Louise wins the first game. (4)

(Total 14 marks)

Q20.

A company analyses the servicing of machinery at one of its large factories.

When a machine is serviced:

B is the event 'the bearings need replacement';

C is the event 'the compressor pump needs replacement';

S is the event 'the solenoid needs replacement'.

The events B, C and S are independent.

B' is the event 'not B'.

C' is the event 'not C'.

A machine is selected at random for servicing.

You are given that $P(B) = 0.1$ and $P(B \cup C) = 0.28$.

(a) (i) Show that $P(C) = 0.2$.

(4)

(ii) Find the value of $P(C' \cup B')$.

(2)

(b) The probability that the solenoid will need replacement is 0.3.

Find the probability that:

(i) the bearings will need replacement, but neither the compressor pump nor the solenoid will need replacement;

(2)

(ii) exactly **one** of the three components will need replacement;

(3)

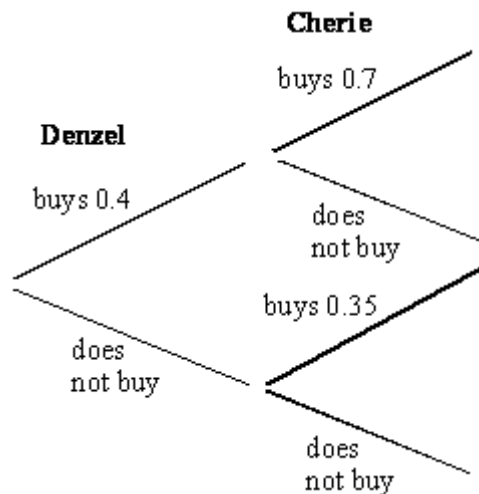
(iii) the solenoid will need replacement, given that exactly one of the three components needs replacement.

(3)

(Total 14 marks)

Q21.

Denzel and Cherie are friends who often go to the cinema together. On such visits there is a probability of 0.4 that Denzel will buy popcorn. The probability that Cherie will buy popcorn is 0.7 if Denzel buys popcorn and 0.35 if he does not.



- (a) When Denzel and Cherie visit the cinema together:
- (i) find the probability that both buy popcorn; (2)
 - (ii) show that the probability that neither buys popcorn is 0.39; (2)
 - (iii) find the probability that exactly one of them buys popcorn. (2)
- (b) Sohaib sometimes joins Denzel and Cherie on their cinema visits. On these occasions, the probability that Sohaib buys popcorn is 0.55 if both Denzel and Cherie buy popcorn and 0.25 if exactly one of Denzel and Cherie buys popcorn.
- Find the probability that when Denzel, Cherie and Sohaib visit the cinema together:
- (i) all three buy popcorn; (2)
 - (ii) Cherie and Sohaib buy popcorn but Denzel does not. (2)
- (Total 10 marks)**

Q22.

At a university, 60% of students are studying Arts subjects and the rest are studying Science subjects. Of the Arts students, 55% are female and of the Science students, 35% are female.

Find the probability that a student, selected at random:

- (a) is female and studying an Arts subject; (2)
- (b) is male and studying a Science subject; (2)

(c) is male; (3)

(d) is studying Science, given that the student is male. (3)

(Total 10 marks)

Q23.

At a university, 60% of students are studying Arts subjects and the rest are studying Science subjects.

Of the Arts students, 55% are female and of the Science students, 35% are female.

(a) Find the probability that a student, selected at random:

(i) is female and studying an Arts subject; (2)

(ii) is male and studying a Science subject; (2)

(iii) is male; (2)

(iv) is studying Science, given that the student is male. (2)

(b) Of all the female students, one third of those studying a Science subject and one half of those studying an Arts subject live in a Hall of Residence. Find the probability that a female student, who lives in a Hall of Residence, is studying a Science subject. (4)

(Total 12 marks)

Q24.

A rugby club has three categories of membership: adult, social and junior. The number of members in each category, classified by gender, is shown in the table below.

	Adult	Social	Junior
Female	25	35	40
Male	95	25	80

One member is chosen, at random, to cut the ribbon at the opening of the new clubhouse.

(a) Find the probability that:

(i) a female member is chosen;

- (ii) a junior member is chosen;
- (iii) a junior member is chosen, given that a female member is chosen.

(4)

- (b) V denotes the event that a female member is chosen.
W denotes the event that an adult member is chosen.
X denotes the event that a junior member is chosen.

For the events V, W and X:

- (i) write down two which are mutually exclusive;
- (ii) find two which are neither mutually exclusive nor independent. Justify your answer.

(1)

(3)

(Total 8 marks)

Q25.

Ricky, Saeed and Tom like to go to the park after school on Wednesday to play football.

The independent probabilities that Ricky, Saeed and Tom go to the park to play football on Wednesday are 0.8, 0.7 and 0.9 respectively.

Find the probability that, on a particular Wednesday:

- (a) none of the three boys goes to the park;
- (b) Ricky is the only one to go to the park;
- (c) only **one** of the three boys goes to the park;
- (d) Ricky goes, given that only one of the three boys goes to the park.

(2)

(2)

(3)

(3)

(Total 10 marks)

Q26.

Keith, Yasmin and Suzie are friends who often go to a jazz club on a Friday night. The probability that Keith goes to the club on a particular Friday is 0.7 and is independent of whether or not Yasmin and Suzie go.

The probability that Yasmin goes to the club is 0.8 if Keith goes and 0.6 if Keith does not go.

The probability that Suzie goes to the club:

- is 0.9 if both Keith and Yasmin go;
- is 0.65 if Keith goes but Yasmin does not;

is 0.55 if Yasmin goes but Keith does not;
is 0.4 if neither Keith nor Yasmin go.

By drawing a tree diagram, or otherwise, find the probability that on a particular Friday:

- (a) all three friends go to the club; (3)
 - (b) Keith goes to the club but Yasmin and Suzie do not; (2)
 - (c) Yasmin and Suzie go to the club but Keith does not; (2)
 - (d) Suzie goes to the club. (3)
- (Total 10 marks)**

Q27.

The senior driving examiner at a test centre decides to collect some data about the candidates who are taking a driving test for the first time. The candidates are asked how many driving lessons they had before they took the test. The results are given in the table below.

Number of lessons	Gender	
	Male	Female
20 or fewer	35	47
between 21 and 50	87	75
more than 50	8	21

A candidate is selected at random.

- M is the event “the candidate is male”.
- T is the event “the candidate had 20 or fewer lessons”.
- S is the event “the candidate had more than 50 lessons”.
- M' is the event “not M”.
- S' is the event “not S”.

- (a) Find:
 - (i) $P(M \cap S)$; (1)
 - (ii) $P(T)$; (1)
 - (iii) $P(M | T)$; (2)

(iv) $P(M' \cap T)$;

(2)

(v) $P(M' | S')$.

(3)

(b) Define the event $M' | S'$ in words as simply as possible.

(2)

(Total 11 marks)

Q28.

Amy, Bruce and Carmen take part in an archery competition. The final test of the competition involves these three archers trying to hit the target centre.

The independent probabilities that Amy, Bruce and Carmen hit the target centre are 0.2, 0.3 and 0.6 respectively.

Find the probability that the target centre will be hit by:

(a) Bruce only;

(2)

(b) Carmen only;

(2)

(c) exactly one of the three archers;

(3)

(d) Bruce, given that exactly one of the three archers hits the target centre.

(3)

(Total 10 marks)

Q29.

A blood test for determining the type of hepatitis from which a patient is suffering involves testing for the Australian antigen. A “positive” result to this test indicates that a patient has hepatitis C.

Unfortunately, this test is not completely reliable.

If an individual has hepatitis C, there is a probability of 0.8 that the test will give a “positive” result. If a patient does **not** have hepatitis C, there is a probability of 0.1 that the test will give a “positive” result.

It is known that 18% of patients suffering from hepatitis have type C.

Find the probability that a person chosen at random from all hepatitis sufferers:

(a) has type C and gives a “positive” result to the test;

(2)

(b) gives a “positive” result to the test;

(3)

(c) does not have hepatitis C, given that the test has given a “positive” result.

(3)

(Total 8 marks)

Q30.

A random sample of 170 students aged between 16 and 21 years was asked their opinions regarding the level of the student loan available to students in higher education.

They were asked to comment on whether they felt the level of the loan was too low, about right or too high.

The following table summarises their replies.

Reply Age of student	Too low	About right	Too high
16 – 17 years	45	17	8
18 – 21 years	40	35	25

A student is chosen at random.

B is the event “the student is aged 16–17 years”.

C is the event “the student replied about right”.

D is the event “the student replied too high”.

(a) Find:

(i) $P(C)$;

(2)

(ii) $P(B \cap D)$;

(1)

(iii) $P(C \cup B)$;

(2)

(iv) $P(D | B)$.

(3)

(b) Define in words the event $B \cup D$.

(2)

(Total 10 marks)

Q31.

A random sample of students aged between 16 and 21 years was asked their opinions regarding the level of the student loan available to students in higher education.

They were asked to comment on whether they felt the level of the loan was too low, about right or too high.

The following table summarises their replies.

	Reply	Too low	About right	Too high
Age of student				
16 – 17 years		45	17	8
18 – 21 years		40	35	25

A student is chosen at random.

B is the event “the student is aged 16–17 years”.

C is the event “the student replied about right”.

D is the event “the student replied too high”.

B' is the event “not B ”.

D' is the event “not D ”.

(a) Find:

(i) $P(B \cap C)$;

(1)

(ii) $P(D \cup B)$;

(2)

(iii) $P(C | B)$;

(2)

(iv) $P(D' \cup B')$.

(2)

(b) Define, in words as simply as possible, the event $(D' \cup B')$.

(2)

(Total 9 marks)

Q32.

A bottle bank consists of three containers, one each for green, brown and clear bottles. A notice requests people using the bottle bank to place bottles in the appropriate container. However some mistakes are made. The following table summarises where the last 200 bottles taken to the bank were placed. For example 15 brown bottles were incorrectly

placed in the container for green bottles.

		Colour of bottle		
		Green	Brown	Clear
Container	Green	85	15	5
	Brown	4	45	6
	Clear	1	5	34

- (a) One of the 200 bottles is chosen at random.

Find the probability that the chosen bottle is:

- (i) green and correctly placed in the green container;
- (ii) placed in the correct container;
- (iii) placed in the correct container given that the bottle is clear.

(5)

- (b) Three of the 200 bottles are chosen at random without replacement.

Find the probability that:

- (i) the first bottle chosen is green;
- (ii) all are green
- (iii) two are green and one is brown.

(8)

(Total 13 marks)